
EM 655: Hydrogen Storage and Production Systems

Course Code	EM 655
Course Name	Hydrogen Storage and Production Systems
L-T-P-C	3-0-0-3
Intended for	M.Tech Energy Engineering
Distribution	Specialisation elective
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Course Modules

- **Module 1 (Production of Hydrogen):** Different methods of hydrogen production, Renewable electrolysis, Steam Methane Reforming, Gasification of coal and other hydrocarbon, Hydrogen from bio-mass, Thernuclear/other methods. The course will include at least few lectures of elaborate study and description about each of the method of hydrogen production. (14)
- **Module 2 (Hydrogen Handling and Transport):** Major issues regarding the handling of hydrogen and safety requirement. Materials issues: hydrogen embrittlement, Hydrogen leakage and monitoring, Hydrogen pressurizing and compression, Liquid hydrogen: liquefaction costs and low temperature, Pipelines for hydrogen transport-main issue, Few lectures will be given on these above issues with detailed discussion. Emphasis will be given on the role of materials science in the improvement of the handling of hydrogen. (12)
- **Module 3 (Hydrogen Storage):** Hydrogen storage requirements: what is crucial? Metal Hydride requirements for hydrogen storage for mobile application, Onboard hydrogen storage system for light-duty vehicles, Hydrogen storage: materials point of view, Storage through surface adsorption: nanocarbon, metal organic framework, Conventional metal hydrides, Light metal hydrides, Clathrates for hydrogen storage, Material Scientist's point of view: scope of research on the materials for hydrogen storage, their stability under loading and unloading of hydrogen. (14)
- **Module 4 (Hydrogen R & D Opportunities):** Production: alternate methods of producing hydrogen, Delivery: methods of producing hydrogen on demands, Storage: Safe handling and storage, Conversion: conversion hydrogen to energy, Applications: methods for hydrogen consumption, Codes and safety: hydrogen technology standards (8)

Text Books

1. Ram B Gupta, *Hydrogen Fuel Production, Transport and Storage*, CRC Press, 2008.